

Soham Rohit Chitnis

New York City | sc11537@nyu.edu | +1 (201) 920-3972 | linkedin.com/in/soham-chitnis | soham-chitnis10.github.io/

Education

New York University, Courant Institute

New York, NY, USA

Master of Science in Computer Science (GPA: **3.96/4.0**)

September 2024 - May 2026

- Courses: Deep Learning, Natural Language Processing (NLP), DevOps, Computer Vision

Birla Institute of Technology and Science Pilani (BITS Pilani)

Goa, India

Bachelor of Engineering in Computer Science, Minor in Data Science

November 2020 – July 2024

- Courses: Machine Learning, Reinforcement Learning, Database Systems, Data Structures & Algorithms

Experience

CILVR Lab, New York University

New York, NY, USA

Graduate Student Researcher

January 2025 – Present

- **Advisors:** Wancong (Kevin) Zhang, Prof. Yann LeCun
- Training world models (JEPA & Hierarchical JEPA) using self-supervised learning for robotic control and planning.

APP Center for AI Research, BITS Pilani

Goa, GA, India

Undergraduate Student Researcher

February 2022 – May 2024

- **Advisors:** Prof. Ashwin Srinivasan, Prof. Tanmay Verlekar
- Proposed a domain-specific model for feature extraction in whole slide image classification using PyTorch, which achieved 96.85% AUC, improved diagnostic confidence, and was published and presented orally at IEEE EMBC
- Developed AutoRef (with TCS Research), an autonomous peer-review multi-agent system where interacting agents generate and critique reviews based on ICLR standards using LangChain and OpenAI API, achieving a 217% improvement in review quality.

Tata Consultancy Services Research

Pune, MH, India

Research Intern

September 2023 – December 2023

- **Advisor:** Dr. Manasi Patwardhan
- Engineered an efficient, plug-and-play multimodal model to visually ground a frozen OPT-125M LLM for chart understanding, using only a trainable linear adapter, fine-tuned CLIP with Charts, and benchmarked vision encoders (CLIP, DePlot) on ChartQA.

VC Lab, Ontario Tech University

Oshawa, ON, Canada

MITACS Globalink Research Intern

June 2023 – August 2023

- **Advisor:** Prof. Faisal Qureshi
- Proposed a Spatial-Attention CNN Latent Dirichlet VAE for Hyperspectral Unmixing during the fully-funded MITACS Globalink Research Internship, reducing estimation standard deviation by 95% and earning an oral presentation at IEEE IGARSS 2024.

Skills

Languages: Python, C++, C, Java, JavaScript, Bash, SQL **Web & API Development:** REST APIs, FastAPI, HTML/CSS
Machine Learning: PyTorch, TensorFlow, JAX, Scikit-Learn, Pandas, NumPy, HuggingFace, LangChain, DSPy, OpenCV, MLFlow
Developer Tools & DevOps: GIT, Docker, CI/CD, AWS, Linux, Slurm, Prefect
Software & Scientific Computing: MATLAB, LaTeX, MS Office Suite (Excel, Word, PowerPoint)

Selected Projects

Self-supervised learning with Latent Space World Model (JEPA)

[Demo](#)

- Engineered JEPA using Barlow Twins with ResNet-18 Context Encoder and LSTM Cell as Predictor under Markov Assumption. Achieved **3rd Rank** in the Competition with a state-of-the-art probing loss of 0.84, which can accurately predict trajectories.

Multi-Task fine-tuning for Machine Translation of Code-Mixed Languages

[Poster](#)

- Trained Llama 3.2 1B with **QLoRA** using **Multi-Task Fine-tuning** to translate Hindi-English sentences to English and Hindi. Improved BLEU scores by 7.82% & 5.32% for English and Romanized Hindi, respectively, in comparison with SFT.

Action-aware Representation Alignment (AC-REPA) for Navigation World Models

[Code](#) [Report](#)

- Trained Action Conditioned Diffusion Transformer (CDiT-S) with AC-REPA loss that improves rollout dynamics by 3.9%, semantic similarity at long horizon by 4.5%, and yields closer temporal smoothness to groundtruth by 6.2% with reduced variance by 32%.

Publications

- AutoRef: Generating Refinements of Reviews Given Guidelines
Chitnis S., Patwardhan, M., Srinivasan, A., Verlekar, T.T., Vig, L., Shroff, G. SDP Workshop, ACL 2024
- SpACNN-LDVAE: Spatial Attention Convolutional Latent Dirichlet Variational Autoencoder for Hyperspectral Pixel Unmixing
Chitnis S., Mantripragada, K., Qureshi, F.Z. **(Oral)** IEEE IGARSS 2024
- Domain-Specific Pre-training Improves Confidence in Whole Slide Image Classification
Chitnis S.R., Liu, S., Dash, T., Verlekar, T.T., Di Ieva, A., Berkovsky, S., Vig, L., Srinivasan, A. **(Oral)** IEEE EMBC 2023